

Enterprise Gym Platform

API Reference

Complete API specification for the fully customizable enterprise gym SaaS platform. Covers Auth, Members, Workouts, Billing, AI/ML, Analytics, Automation, Multi-Tenant, Integrations, Gamification, Realtime, and more.

15

API Domains

90+

API Groups

400+

Endpoints

REST · GraphQL · WebSocket · OpenAPI 3.0 Compatible

01. AUTHENTICATION & IDENTITY

Login, token management, SSO, face recognition, biometric access, and session control.

Authentication API

10 endpoints

Core login, logout, token refresh, and password management flows.

POST	/auth/register	Register new user account
POST	/auth/login	Authenticate and receive JWT tokens
POST	/auth/logout	Invalidate current session tokens
POST	/auth/refresh-token	Renew access token using refresh token
POST	/auth/forgot-password	Initiate password reset flow
POST	/auth/reset-password	Complete password reset with token
PUT	/auth/change-password	Update password for authenticated user
GET	/auth/me	Retrieve current authenticated user profile
POST	/auth/verify-email	Verify email address via OTP
POST	/auth/resend-otp	Resend email or SMS OTP

SSO & OAuth API

5 endpoints

Single Sign-On and third-party OAuth integrations.

POST	/auth/sso/init	Initiate SSO with provider (Google, Microsoft, Apple)
POST	/auth/sso/callback	Handle OAuth2 callback and issue platform tokens
GET	/auth/sso/providers	List configured SSO providers for tenant
PUT	/auth/sso/providers/:id	Configure SSO provider settings
DELETE	/auth/sso/providers/:id	Remove SSO provider configuration

Biometric Entry API

6 endpoints

Face recognition and biometric device access control for gym entry.

POST	/biometric/face/enroll	Enroll member face data for recognition
POST	/biometric/face/verify	Verify identity via live face scan
DELETE	/biometric/face/:userId	Remove stored biometric data
GET	/biometric/entry/logs	Retrieve gym entry event logs
POST	/biometric/fingerprint/enroll	Enroll fingerprint for access
POST	/biometric/fingerprint/verify	Verify fingerprint at entry point

Roles & Permissions API

11 endpoints

Custom role creation, permissions matrix, and user role assignments.

GET	/roles	List all roles in tenant
POST	/roles	Create a new custom role
GET	/roles/:id	Get role details and permission set
PUT	/roles/:id	Update role name and description
DELETE	/roles/:id	Delete a custom role
GET	/roles/:id/permissions	List all permissions assigned to role
PUT	/roles/:id/permissions	Replace full permissions set for role
GET	/permissions	List all available permissions
GET	/permissions/matrix	Get full permission matrix for all roles
POST	/users/:id/roles	Assign role to a user
DELETE	/users/:id/roles/:roleId	Revoke role from a user

02. USER & MEMBER MANAGEMENT

Members, trainers, leads, staff, onboarding, assessments, and corporate accounts.

Members API

15 endpoints

Full lifecycle management for gym members.

GET	/members	List members with filters and pagination
POST	/members	Create new member profile
GET	/members/:id	Get member profile details
PUT	/members/:id	Update member profile information
DELETE	/members/:id	Soft delete / archive member
POST	/members/bulk	Bulk import members via CSV/JSON
GET	/members/:id/timeline	Get full activity timeline for member
POST	/members/:id/photo	Upload member profile photo
GET	/members/:id/notes	Retrieve trainer notes for member
POST	/members/:id/notes	Add trainer note to member
GET	/members/:id/documents	List member documents and consents
POST	/members/:id/documents	Upload document for member
GET	/members/search	Full-text search across member profiles
GET	/members/:id/tags	List tags assigned to member
POST	/members/:id/tags	Assign tags to member

Trainers API

12 endpoints

Trainer profiles, client assignments, schedules, and performance.

GET	/trainers	List all trainers
POST	/trainers	Create trainer profile
GET	/trainers/:id	Get trainer details
PUT	/trainers/:id	Update trainer profile
GET	/trainers/:id/clients	List clients assigned to trainer
POST	/trainers/:id/assign	Assign member to trainer
DELETE	/trainers/:id/clients/:cid	Unassign member from trainer
GET	/trainers/:id/schedule	Get trainer availability schedule
PUT	/trainers/:id/schedule	Set trainer availability slots
GET	/trainers/:id/performance	Get trainer performance analytics
GET	/trainers/:id/earnings	Get trainer earnings/commission data
POST	/trainers/auto-assign	Auto-assign trainer based on AI matching

Leads API

9 endpoints

Lead capture, scoring, nurturing, and conversion tracking.

GET	/leads	List all leads with status filters
POST	/leads	Create a new lead record
GET	/leads/:id	Get lead details
PUT	/leads/:id	Update lead information
POST	/leads/:id/convert	Convert lead to active member
GET	/leads/:id/score	Get AI-generated lead score
POST	/leads/:id/followup	Schedule or log a follow-up
GET	/leads/sources	Get lead source breakdown analytics
POST	/leads/import	Bulk import leads from CSV

Onboarding API

7 endpoints

Guided onboarding flows, assessments, and fitness goal setup.

POST	/onboarding/start	Initialize onboarding flow for new member
GET	/onboarding/:id/status	Get current onboarding step and completion
PUT	/onboarding/:id/step	Submit data for current onboarding step
POST	/onboarding/assessments	Submit initial fitness assessment
GET	/onboarding/templates	List onboarding flow templates
POST	/onboarding/templates	Create custom onboarding template
POST	/onboarding/:id/complete	Mark onboarding as complete

Corporate Accounts API

8 endpoints

Corporate membership tiers, employee management, and billing.

GET	/corporate/accounts	List all corporate accounts
POST	/corporate/accounts	Create new corporate account
GET	/corporate/accounts/:id	Get corporate account details
PUT	/corporate/accounts/:id	Update corporate account settings
GET	/corporate/accounts/:id/members	List employees under corporate account
POST	/corporate/accounts/:id/members	Add employee to corporate account
DELETE	/corporate/accounts/:id/members/:uid	Remove employee from account
GET	/corporate/accounts/:id/billing	Get corporate billing summary

03. WORKOUTS & EXERCISE ENGINE

Workout builder, exercise library, plans, progression trees, circuits, and AI generation.

Exercise Library API

11 endpoints

CRUD for exercises, muscle groups, video, tags, and custom exercises.

GET	/exercises	List exercises with tag/muscle filters
POST	/exercises	Create custom exercise
GET	/exercises/:id	Get exercise details
PUT	/exercises/:id	Update exercise metadata
DELETE	/exercises/:id	Delete custom exercise
GET	/exercises/:id/alternatives	Get injury-safe alternatives
POST	/exercises/:id/video	Upload exercise demo video
POST	/exercises/:id/video/annotate	Add coaching annotations to video
GET	/exercises/tags	List all exercise tags
GET	/exercises/muscles	List all muscle group categories
GET	/exercises/equipment	List equipment-filtered exercises

Workouts API

12 endpoints

Create, assign, clone, and track individual workout sessions.

GET	/workouts	List workouts with filters
POST	/workouts	Create new workout
GET	/workouts/:id	Get workout details with exercises
PUT	/workouts/:id	Update workout
DELETE	/workouts/:id	Delete workout
POST	/workouts/:id/clone	Clone workout template
POST	/workouts/assign	Assign workout to member(s)
GET	/workouts/:id/progress	Get member progress on workout
POST	/workouts/:id/complete	Log workout completion with stats
GET	/workouts/:id/score	Get performance score for workout
POST	/workouts/:id/share	Share workout to social feed
POST	/workouts/:id/bookmark	Bookmark workout for member

Workout Plans API

11 endpoints

Multi-week progressive plans, branching logic, and progression trees.

GET	/plans	List workout plans
POST	/plans	Create new workout plan
GET	/plans/:id	Get plan with full schedule
PUT	/plans/:id	Update plan settings
DELETE	/plans/:id	Delete plan
GET	/plans/:id/tree	Get progression tree visualization data
POST	/plans/:id/branch	Add conditional branch to plan
PUT	/plans/:id/progression	Update progression rules
POST	/plans/:id/assign	Assign plan to member(s)
GET	/plans/:id/members	List members on this plan
GET	/plans/:id/analytics	Get plan completion analytics

Workout Builder Components API

9 endpoints

Circuits, supersets, drop-sets, pyramids, and timer configurations.

POST	/workouts/:id/circuits	Add circuit block to workout
PUT	/workouts/:id/circuits/:cid	Update circuit settings
POST	/workouts/:id/supersets	Add superset block
POST	/workouts/:id/dropsets	Add drop-set configuration
POST	/workouts/:id/pyramids	Add pyramid set configuration
PUT	/workouts/:id/tempo	Set tempo control for exercises
PUT	/workouts/:id/rest-intervals	Configure rest interval rules
POST	/workouts/:id/timer	Configure workout timer engine
PUT	/workouts/:id/difficulty	Set auto-adjustment difficulty rules

Workout Automation API

6 endpoints

Rules-based automated workout assignment and progression.

GET	/workout-automation/rules	List workout automation rules
POST	/workout-automation/rules	Create automation rule
PUT	/workout-automation/rules/:id	Update automation rule
DELETE	/workout-automation/rules/:id	Delete automation rule
POST	/workout-automation/trigger	Manually trigger automation rule
GET	/workout-automation/logs	Get automation execution logs

04. MEMBERSHIP & BILLING

Membership plans, subscriptions, payments, invoicing, dynamic pricing, and revenue analytics.

Membership Plans API

7 endpoints

Plan definition, feature unlocking, and subscription tiers.

GET	/membership-plans	List all available membership plans
POST	/membership-plans	Create new membership plan
GET	/membership-plans/:id	Get plan details and features
PUT	/membership-plans/:id	Update plan settings
DELETE	/membership-plans/:id	Archive membership plan
GET	/membership-plans/:id/features	List features unlocked by plan
PUT	/membership-plans/:id/features	Update plan feature set

Subscriptions API

10 endpoints

Member subscription lifecycle, renewals, freezes, and upgrades.

GET	/subscriptions	List all member subscriptions
POST	/subscriptions	Create new subscription for member
GET	/subscriptions/:id	Get subscription details
PUT	/subscriptions/:id/renew	Renew expiring subscription
POST	/subscriptions/:id/upgrade	Upgrade to higher tier plan
POST	/subscriptions/:id/downgrade	Downgrade subscription tier
POST	/subscriptions/:id/freeze	Freeze subscription for a period
POST	/subscriptions/:id/unfreeze	Reactivate frozen subscription
DELETE	/subscriptions/:id/cancel	Cancel subscription
GET	/subscriptions/:id/usage	Get usage metrics for subscription

Payments API

9 endpoints

Payment processing, refunds, invoices, and payment history.

POST	/payments/charge	Process a payment
GET	/payments/:id	Get payment details
POST	/payments/refund	Process full or partial refund
GET	/payments/history	List payment history with filters
POST	/payments/methods	Save new payment method
GET	/payments/methods	List saved payment methods
DELETE	/payments/methods/:id	Remove saved payment method
POST	/payments/intent	Create Stripe/Razorpay payment intent
POST	/payments/reminders	Send manual payment reminder

Invoices API

6 endpoints

Invoice generation, PDF export, and accounting integration.

GET	/invoices	List invoices with filters
POST	/invoices	Generate invoice manually
GET	/invoices/:id	Get invoice details
GET	/invoices/:id/pdf	Download invoice as PDF
POST	/invoices/:id/send	Email invoice to member
PUT	/invoices/:id/mark-paid	Mark invoice as manually paid

Dynamic Pricing API

8 endpoints

Time-based, demand-based, and promotional pricing rules.

GET	/pricing/rules	List all dynamic pricing rules
POST	/pricing/rules	Create new pricing rule
PUT	/pricing/rules/:id	Update pricing rule
DELETE	/pricing/rules/:id	Delete pricing rule
GET	/pricing/calculate	Calculate price for given context
GET	/pricing/discounts	List active discount codes
POST	/pricing/discounts	Create discount/promo code
POST	/pricing/discounts/validate	Validate discount code at checkout

05. SCHEDULING & ATTENDANCE

Class scheduling, session bookings, attendance tracking, and access control.

Classes & Sessions API

12 endpoints

Class creation, scheduling, capacity, and booking management.

GET	/classes	List classes with date/trainer filters
POST	/classes	Create new class or session
GET	/classes/:id	Get class details
PUT	/classes/:id	Update class settings
DELETE	/classes/:id	Cancel class
POST	/classes/:id/book	Book spot in class for member
DELETE	/classes/:id/book/:uid	Cancel member booking
GET	/classes/:id/roster	Get confirmed attendees list
GET	/classes/:id/waitlist	Get waitlist for full class
POST	/classes/:id/waitlist	Join waitlist for full class
POST	/classes/recurring	Create recurring class series
GET	/classes/calendar	Get monthly calendar view data

Attendance API

9 endpoints

Check-in, check-out, history, and heatmap analytics.

POST	/attendance/check-in	Log member check-in event
POST	/attendance/check-out	Log member check-out event
GET	/attendance/history	Get attendance history with filters
GET	/attendance/today	Get today's live attendance count
GET	/attendance/heatmap	Get hourly attendance heatmap data
GET	/attendance/:userId	Get attendance history for member
POST	/attendance/qr-scan	Process QR code check-in
GET	/attendance/analytics	Get attendance analytics report
GET	/attendance/predictions	AI-predicted attendance for next 7 days

Personal Training Sessions API

9 endpoints

PT session scheduling, notes, and billing integration.

GET	/pt-sessions	List PT sessions
POST	/pt-sessions	Book new PT session
GET	/pt-sessions/:id	Get PT session details
PUT	/pt-sessions/:id	Update PT session
DELETE	/pt-sessions/:id	Cancel PT session
POST	/pt-sessions/:id/complete	Mark PT session complete
POST	/pt-sessions/:id/notes	Add session notes
GET	/pt-sessions/packages	List PT session packages
POST	/pt-sessions/packages	Create PT session package

06. AI & MACHINE LEARNING

AI workout generation, diet planning, predictions, pose estimation, chatbot, and smart recommendations.

AI Workout Generation API

6 endpoints

Generate, optimize, and substitute workouts using ML models.

POST	/ai/workouts/generate	Generate personalized workout from member profile
POST	/ai/workouts/optimize	Optimize existing workout for member goals
POST	/ai/workouts/substitute	Get AI exercise substitutions (injury-aware)
POST	/ai/workouts/difficulty	Auto-adjust workout difficulty via ML
GET	/ai/workouts/recommendations/:uid	Get workout recommendations for member
POST	/ai/workouts/fatigue-check	Evaluate fatigue level before assigning workout

AI Diet & Nutrition API

6 endpoints

Personalized meal plans, macro calculation, and nutrition recommendations.

POST	/ai/diet/generate	Generate meal plan based on goals and macros
GET	/ai/diet/:userId/plan	Get current AI diet plan for member
PUT	/ai/diet/:userId/preferences	Update dietary restrictions and goals
POST	/ai/diet/analyze	Analyze uploaded meal photo for macros
GET	/ai/diet/alternatives	Get food alternative suggestions
POST	/ai/diet/log	Log meal intake for tracking

AI Predictions API

7 endpoints

Churn prediction, attendance forecast, transformation prediction, and injury risk.

POST	/ai/predict/churn	Predict churn probability for member
GET	/ai/predict/churn/bulk	Bulk churn risk scores for all members
POST	/ai/predict/attendance	Predict attendance for time slots
POST	/ai/predict/transformation	Predict body transformation timeline
GET	/ai/predict/revenue	Revenue forecast for next 90 days
POST	/ai/predict/injury-risk	Evaluate injury risk from workout load
GET	/ai/leads/score/:id	Get ML-based lead conversion score

Pose Estimation & Posture API

6 endpoints

Realtime pose detection, posture correction, and form scoring.

POST	/ai/pose/estimate	Run pose estimation on uploaded video frame
POST	/ai/posture/analyze	Analyze posture from image/video
GET	/ai/posture/corrections/:id	Get posture correction history for member
POST	/ai/posture/realtime	Start realtime posture monitoring session
POST	/ai/form/score	Score exercise form from video
POST	/ai/injury/detect	Detect potential injury indicators in movement

AI Chatbot & Voice API

7 endpoints

Conversational trainer bot, voice coaching commands, and support routing.

POST	/ai/chat	Send message to AI trainer chatbot
GET	/ai/chat/history/:userId	Retrieve conversation history
DELETE	/ai/chat/history/:userId	Clear conversation history
POST	/ai/voice/transcribe	Transcribe voice command to text
POST	/ai/voice/coach	Get voice coaching response for action
POST	/ai/support/route	Route support ticket via AI classification
GET	/ai/support/suggestions/:id	Get AI response suggestions for ticket

AI Analytics & Reports API

6 endpoints

AI-generated insight summaries, business intelligence, and automated reports.

POST	/ai/reports/generate	Generate AI-written analytics report
GET	/ai/insights/business	Get AI business health insights
GET	/ai/insights/members	Get AI member engagement insights
POST	/ai/search	AI-powered natural language search
GET	/ai/personalization/:uid	Get personalization profile for member
PUT	/ai/personalization/:uid	Update personalization preferences

07. HEALTH & WELLNESS TRACKING

Goals, habits, mood, sleep, hydration, recovery, stress, and wearable device sync.

Goals API

6 endpoints

Custom fitness goal creation, tracking, and milestone management.

GET	/goals/:userId	Get all active goals for member
POST	/goals	Create new fitness goal
PUT	/goals/:id	Update goal target or deadline
DELETE	/goals/:id	Remove goal
POST	/goals/:id/progress	Log progress entry toward goal
GET	/goals/:id/history	Get historical progress for goal

Daily Tracking API

9 endpoints

Habits, mood, hydration, sleep, stress, and recovery logging.

POST	/tracking/habits	Log daily habit completion
GET	/tracking/habits/:userId	Get habit streak and history
POST	/tracking/mood	Log daily mood entry
POST	/tracking/sleep	Log sleep quality and duration
POST	/tracking/hydration	Log water intake
POST	/tracking/stress	Log stress level entry
POST	/tracking/recovery	Log recovery metrics
GET	/tracking/dashboard/:userId	Get aggregated daily tracking dashboard
GET	/tracking/trends/:userId	Get 30-day health trends

Transformation Journal API

6 endpoints

Progress photos, body measurements, and transformation timeline.

POST	/journal/photos	Upload progress photo
GET	/journal/photos/:userId	Get progress photo gallery
DELETE	/journal/photos/:id	Delete progress photo
POST	/journal/measurements	Log body measurements
GET	/journal/measurements/:uid	Get measurement history
GET	/journal/timeline/:userId	Get full transformation timeline

Wearables & Devices API

10 endpoints

Connect wearables, sync health data, and manage device integrations.

GET	/devices	List connected devices for user
POST	/devices/connect	Connect a wearable device
DELETE	/devices/:id	Disconnect device
POST	/devices/sync	Trigger manual data sync
GET	/devices/:id/data	Get synced health data from device
POST	/integrations/google-fit	Sync data from Google Fit
POST	/integrations/apple-health	Sync data from Apple Health
POST	/integrations/garmin	Sync data from Garmin Connect
POST	/integrations/fitbit	Sync data from Fitbit API
POST	/integrations/whoop	Sync recovery data from WHOOP

08. ANALYTICS & REPORTING

Realtime dashboards, revenue analytics, member engagement, branch comparisons, and custom reports.

Revenue Analytics API

7 endpoints

Revenue tracking, forecasting, subscription growth, and financial KPIs.

GET	/analytics/revenue	Get revenue summary with date range
GET	/analytics/revenue/forecast	Get 90-day revenue forecast
GET	/analytics/revenue/breakdown	Get revenue by plan, branch, and source
GET	/analytics/subscriptions/growth	Get subscription growth over time
GET	/analytics/ltv	Get member lifetime value metrics
GET	/analytics/mrr	Get monthly recurring revenue data
GET	/analytics/churn-rate	Get churn rate trends

Member Engagement Analytics API

6 endpoints

Engagement scores, retention metrics, activity analysis, and behavioral insights.

GET	/analytics/engagement	Get member engagement scores
GET	/analytics/retention	Get retention rates by cohort
GET	/analytics/conversion	Get lead-to-member conversion funnel
GET	/analytics/activity	Get workout completion statistics
GET	/analytics/nps	Get Net Promoter Score data
GET	/analytics/segments	Get member segmentation breakdown

Operational Analytics API

6 endpoints

Branch comparisons, trainer productivity, class performance, and equipment usage.

GET	/analytics/branches/compare	Compare KPIs across branches
GET	/analytics/trainers/productivity	Get trainer productivity scores
GET	/analytics/classes/performance	Get class fill rate and satisfaction
GET	/analytics/equipment/usage	Get gym equipment usage stats
GET	/analytics/peak-hours	Get peak usage hour analysis
GET	/analytics/capacity	Get current capacity utilization

Custom Reports API

9 endpoints

Report builder, scheduling, export, and data warehouse integration.

GET	/reports	List saved custom reports
POST	/reports	Create new custom report
GET	/reports/:id	Run and retrieve report data
PUT	/reports/:id	Update report configuration
DELETE	/reports/:id	Delete custom report
POST	/reports/:id/schedule	Schedule automated report delivery
GET	/reports/:id/export	Export report as CSV/Excel/PDF
GET	/reports/audit-trail	Get system audit trail log
GET	/reports/kpi	Get KPI dashboard data

09. NOTIFICATIONS & COMMUNICATIONS

Push notifications, email, SMS, WhatsApp, in-app alerts, templates, and campaign management.

Notifications API

10 endpoints

Send and manage all notification types across channels.

POST	/notifications/send	Send notification via any channel
GET	/notifications/:userId	Get notification inbox for user
PUT	/notifications/:id/read	Mark notification as read
DELETE	/notifications/:id	Delete notification
GET	/notifications/unread/:uid	Get unread count for user
POST	/notifications/push	Send push notification to device(s)
POST	/notifications/email	Send transactional email
POST	/notifications/sms	Send SMS notification
POST	/notifications/whatsapp	Send WhatsApp message
POST	/notifications/in-app	Send in-app notification

Notification Templates API

6 endpoints

Create, manage, and preview notification templates.

GET	/notification-templates	List all notification templates
POST	/notification-templates	Create new notification template
GET	/notification-templates/:id	Get template with variable definitions
PUT	/notification-templates/:id	Update template content
DELETE	/notification-templates/:id	Delete template
POST	/notification-templates/:id/preview	Preview rendered template

Campaigns API

9 endpoints

Email/SMS marketing campaigns, scheduling, and performance tracking.

GET	/campaigns	List all campaigns
POST	/campaigns	Create new campaign
GET	/campaigns/:id	Get campaign details
PUT	/campaigns/:id	Update campaign
POST	/campaigns/:id/launch	Launch campaign immediately
POST	/campaigns/:id/schedule	Schedule campaign for future launch
GET	/campaigns/:id/stats	Get open rates, clicks, conversions
POST	/campaigns/:id/pause	Pause active campaign
GET	/campaigns/segments	List audience segments for targeting

Preferences API

4 endpoints

Manage user notification preferences and opt-outs.

GET	/preferences/:userId	Get notification preferences for user
PUT	/preferences/:userId	Update notification preferences
POST	/preferences/:userId/opt-out	Opt out of a notification type
GET	/preferences/channels	Get available notification channels

10. AUTOMATION & WORKFLOWS

Rule-based automation, event-driven workflows, webhooks, and scheduled tasks.

Automation Rules API

9 endpoints

Create if-then automation rules triggered by events or schedules.

GET	/automations	List all automation rules
POST	/automations	Create new automation rule
GET	/automations/:id	Get automation rule details
PUT	/automations/:id	Update automation rule
DELETE	/automations/:id	Delete automation rule
PUT	/automations/:id/toggle	Enable or disable automation
POST	/automations/trigger	Manually fire an automation
GET	/automations/history	Get automation execution history
GET	/automations/logs/:id	Get detailed execution log

Webhooks API

8 endpoints

Register, manage, and monitor outbound webhook endpoints.

GET	/webhooks	List all registered webhooks
POST	/webhooks	Register new webhook endpoint
GET	/webhooks/:id	Get webhook configuration
PUT	/webhooks/:id	Update webhook endpoint or events
DELETE	/webhooks/:id	Delete webhook registration
POST	/webhooks/:id/test	Send test payload to webhook
GET	/webhooks/:id/logs	Get webhook delivery logs
POST	/webhooks/:id/retry	Retry failed webhook deliveries

Scheduled Tasks API

6 endpoints

Cron-based task scheduling for recurring platform jobs.

GET	/tasks	List all scheduled tasks
POST	/tasks	Create new scheduled task
PUT	/tasks/:id	Update task schedule or action
DELETE	/tasks/:id	Delete scheduled task
POST	/tasks/:id/run	Trigger task execution immediately
GET	/tasks/:id/history	Get task execution history

11. MULTI-BRANCH & TENANT MANAGEMENT

Franchise hierarchy, branch control, tenant isolation, white-labeling, and SaaS admin.

Branches API

10 endpoints

Create and manage gym branches within a franchise or chain.

GET	/branches	List all branches
POST	/branches	Create new branch
GET	/branches/:id	Get branch details
PUT	/branches/:id	Update branch information
DELETE	/branches/:id	Archive branch
GET	/branches/:id/config	Get branch-specific configuration
PUT	/branches/:id/config	Update branch configuration
GET	/branches/:id/analytics	Get analytics for specific branch
GET	/branches/:id/staff	List staff assigned to branch
POST	/branches/:id/staff	Assign staff to branch

Tenants API

9 endpoints

SaaS tenant provisioning, isolation, and subscription management.

GET	/tenants	List all tenants (super admin)
POST	/tenants	Provision new tenant
GET	/tenants/:id	Get tenant details
PUT	/tenants/:id	Update tenant settings
DELETE	/tenants/:id	Suspend tenant
GET	/tenants/:id/usage	Get tenant resource usage metrics
PUT	/tenants/:id/branding	Update white-label branding
PUT	/tenants/:id/features	Toggle feature access for tenant
POST	/tenants/:id/migrate	Migrate tenant to new plan/infrastructure

Platform Configuration API

14 endpoints

Module toggles, feature flags, custom fields, and theme management.

GET	/config/modules	List all platform modules and status
PUT	/config/modules/:id/toggle	Enable or disable a platform module
GET	/config/feature-flags	Get all feature flags for tenant
PUT	/config/feature-flags	Bulk update feature flags
GET	/config/custom-fields	List custom data fields
POST	/config/custom-fields	Create custom field
PUT	/config/custom-fields/:id	Update custom field definition
DELETE	/config/custom-fields/:id	Delete custom field
GET	/config/themes	List available UI themes
PUT	/config/themes/active	Set active theme for tenant
GET	/config/languages	List supported languages
PUT	/config/language	Set active language for tenant
GET	/config/timezone	Get tenant timezone setting
PUT	/config/timezone	Update tenant timezone

12. INTEGRATIONS & EXTERNAL APIS

Payment gateways, CRM, accounting, marketing platforms, IoT equipment, and public developer API.

Payment Gateway API

6 endpoints

Connect and manage payment gateway providers.

GET	/integrations/payment-gateways	List connected gateways
POST	/integrations/payment-gateways	Connect new payment gateway
PUT	/integrations/payment-gateways/:id	Update gateway credentials
DELETE	/integrations/payment-gateways/:id	Disconnect payment gateway
POST	/integrations/stripe/webhook	Stripe event webhook receiver
POST	/integrations/razorpay/webhook	Razorpay event webhook receiver

CRM Integration API

5 endpoints

Sync members and leads with external CRM platforms.

POST	/integrations/crm/connect	Connect CRM platform (HubSpot, Salesforce)
POST	/integrations/crm/sync	Push member/lead data to CRM
GET	/integrations/crm/status	Get last sync status
POST	/integrations/crm/pull	Pull updates from CRM into platform
DELETE	/integrations/crm/disconnect	Disconnect CRM integration

Accounting Integration API

4 endpoints

Sync financial data with accounting platforms.

POST	/integrations/accounting/connect	Connect QuickBooks/Xero/Tally
POST	/integrations/accounting/sync	Sync invoices and payments
GET	/integrations/accounting/status	Get sync status
DELETE	/integrations/accounting/disconnect	Disconnect accounting integration

IoT & Equipment API

7 endpoints

Connect smart gym equipment and biometric reader devices.

GET	/iot/equipment	List connected gym equipment
POST	/iot/equipment/connect	Register new IoT equipment
GET	/iot/equipment/:id/data	Get real-time equipment sensor data
GET	/iot/equipment/:id/usage	Get equipment usage statistics
DELETE	/iot/equipment/:id	Deregister equipment
POST	/iot/biometric/scan	Process biometric scan event
GET	/iot/access-control/logs	Get access control event log

Marketing Platform API

4 endpoints

Sync audiences and events with marketing tools.

POST	/integrations/marketing/connect	Connect Mailchimp/Klaviyo/ActiveCampaign
POST	/integrations/marketing/sync	Sync audience segments
POST	/integrations/marketing/events	Send behavioral events to marketing tool
DELETE	/integrations/marketing/disconnect	Disconnect marketing integration

Public Developer API

7 endpoints

Public REST and GraphQL APIs for third-party developers.

GET	/api/v1/public/members	Public: List members (scoped by API key)
GET	/api/v1/public/classes	Public: List classes and schedules
POST	/api/v1/public/bookings	Public: Book a class or PT session
GET	/api/graphql	GraphQL API endpoint
GET	/api/v1/rate-limits	Get current API rate limit status
POST	/api/keys	Generate new API key
DELETE	/api/keys/:id	Revoke API key

13. GAMIFICATION & SOCIAL

Challenges, achievements, leaderboards, badges, social profiles, and fitness community features.

Challenges API

10 endpoints

Create, manage, and participate in fitness challenges.

GET	/challenges	List active and upcoming challenges
POST	/challenges	Create new challenge
GET	/challenges/:id	Get challenge details
PUT	/challenges/:id	Update challenge settings
DELETE	/challenges/:id	Cancel challenge
POST	/challenges/:id/join	Join a challenge
DELETE	/challenges/:id/leave	Leave a challenge
GET	/challenges/:id/leaderboard	Get challenge leaderboard
POST	/challenges/:id/log	Log progress toward challenge
POST	/challenges/auto-create	Auto-generate AI challenge for members

Achievements & Badges API

7 endpoints

Award badges, track milestones, and manage gamification rewards.

GET	/achievements/:userId	Get all achievements for member
POST	/achievements/award	Manually award achievement to member
GET	/achievements/catalog	List all available achievements
POST	/achievements	Create new achievement definition
PUT	/achievements/:id	Update achievement criteria
DELETE	/achievements/:id	Delete achievement
GET	/badges/:userId	Get badges earned by member

Social Features API

9 endpoints

Social profiles, workout sharing, follows, and activity feeds.

GET	/social/profiles/:id	Get member social fitness profile
PUT	/social/profiles/:id	Update social profile
GET	/social/feed/:userId	Get personalized activity feed
POST	/social/workouts/share	Share workout to social feed
POST	/social/follow/:userId	Follow another member
DELETE	/social/follow/:userId	Unfollow a member
GET	/social/followers/:userId	Get followers list
POST	/social/posts/:id/like	Like a post in the feed
POST	/social/posts/:id/comment	Comment on a post

14. MEDIA & FILE MANAGEMENT

File uploads, video storage, profile photos, documents, and cloud storage management.

Media Upload API

6 endpoints

Upload, manage, and serve media files across the platform.

POST	/media/upload	Upload file (image, video, document)
GET	/media/:id	Get media file metadata and URL
DELETE	/media/:id	Delete media file
GET	/media	List media files with filters
POST	/media/upload/bulk	Bulk file upload
GET	/media/storage/usage	Get cloud storage usage stats

Forms Builder API

8 endpoints

Dynamic form creation, submission handling, and response management.

GET	/forms	List all custom forms
POST	/forms	Create new custom form
GET	/forms/:id	Get form with field definitions
PUT	/forms/:id	Update form structure
DELETE	/forms/:id	Delete form
POST	/forms/:id/submit	Submit form response
GET	/forms/:id/responses	Get all form responses
GET	/forms/:id/responses/:rid	Get specific form response

15. REALTIME & WEBSOCKET

Live coaching streams, dashboard metrics, posture feeds, notifications, and synchronization.

WebSocket Channels

8 endpoints

Persistent realtime connections for live data streams.

WS	/ws/live-coaching	Realtime live PT coaching session
WS	/ws/posture-feed	Realtime posture correction stream
WS	/ws/attendance-feed	Live gym attendance count stream
WS	/ws/dashboard-metrics	Realtime KPI dashboard updates
WS	/ws/notifications	Real-time user notification stream
WS	/ws/leaderboard	Live challenge leaderboard updates
WS	/ws/equipment/:id	Realtime IoT equipment data stream
WS	/ws/workout-session/:id	Live workout progress sync

Realtime Sync API

4 endpoints

REST-based sync endpoints for multi-device and cross-branch synchronization.

GET	/sync/status	Get realtime sync health status
POST	/sync/force	Force sync for a user or branch
GET	/sync/conflicts	Get pending sync conflict records
POST	/sync/resolve	Resolve a data sync conflict